[18MSCH2H02]

Reg. No. 18MSRCH011

SECOND SEMESTER UNIVERSITY EXAMINATION, MAY - 2019

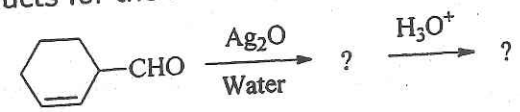
ORGANIC CHEMISTRY - II

Time: 03 Hours | Maximum Marks:70

Instructions: I. Question 1 is compulsory, carries 10 marks.
II. Questions 2 to 7 carry 15 marks each, answer any four.

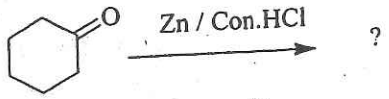
1. Answer any FIVE. 5×2=10

a. Predict the products for the following reactions:

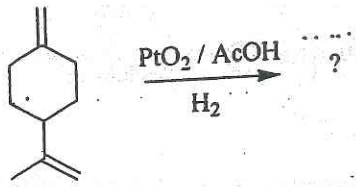


b. Predict the product for the following reactions:

i.



ii.

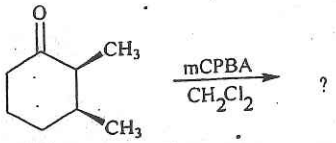


c. Write the two important conditions for the successful synthesis of oxytanes.

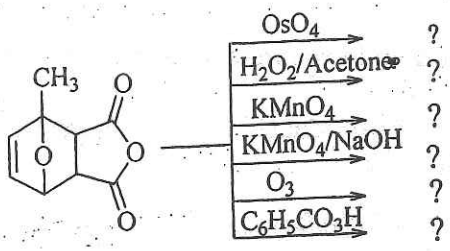
d. Give the photochemical rule for intramolecular dimerization of alkenes

e. Explain the ring opening reactions of oxetane.

f. Predict the product of the following reaction:



2. a. Predict the products and explain the mechanism for the following reactions (any five):



b. Explain the Sharpless assymmetric epoxidation with mechanism. (10 + 5)

3. a. Discuss the mechanistic approach involved in pinacol coupling reaction and Horiuti–Polanyi reduction with suitable examples.

b. Explain the mechanism and enantioselectivity of Noyori asymmetric transfer hydrogenation? (10 + 5)

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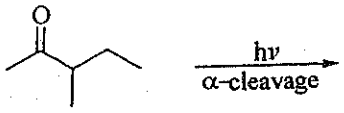
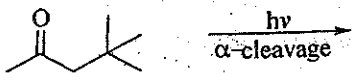
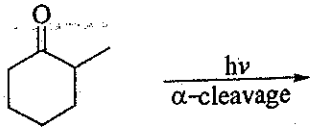
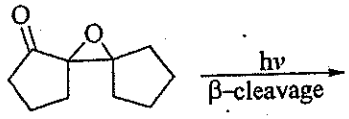
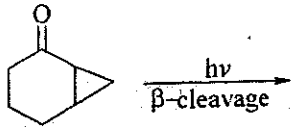
4. a. Predict the products for any four of the following photochemical reactions:

i. 

ii. 

iii. 

iv. 

v. 

b. Differentiate a Norrish type I reaction from a Norrish type II reaction with appropriate examples. **(10 + 5)**

5. a. Describe the mechanism of an intermolecular Paterno-Buchi reaction with an appropriate example. Explain in detail, the influence of an electron rich and electron deficient alkene on Paterno-Buchi reactions by energy level diagrams. Illustrate an intramolecular Paterno-Buchi reaction with an example.

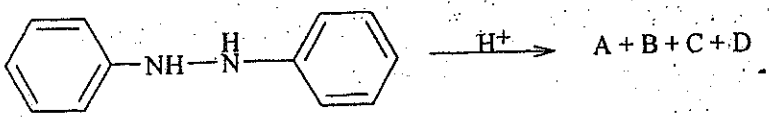
b. Write a note on photo Fries rearrangement. **(10 + 5)**

6. a. i. Write any two methods each for the synthesis of oxetane and thietane

ii. Discuss the ring-opening reactions of thiirane.

b. Explain any three methods for the synthesis of imidazole. **(10 + 5)**

7. a. Predict the products and write the mechanism for following reaction:



b. How do you convert acyl azide to primary amine? **(10 + 5)**